

A Survey of Device Independent Quantum Key Distribution Development

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Abstract—Quantum Key Distribution (QKD) is an emerging technology to mitigate the threat that quantum computers pose to modern key distribution, which relies on the assumption of computational complexity. However, QKD has proved to have shortcomings, including critical vulnerabilities that expose secret keys and compatibility issues between different devices. Device Independent Quantum Key Distribution (DIQKD) is a new approach built on QKD that works by removing assumptions about the communication devices and treating them as black boxes. To operate, the protocol instead relies on performing the Clauser-Horne-Shimony-Holt (CHSH) test to detect Bell-state violations that certify the security of the transmission. In this paper, we review some of the first major development methodologies that have enabled advancements through theoretical and experimental testing. We will examine how Schwonnek et al. established an improvement over the original protocol by introducing two random bases for measurement to reduce critical CHSH scores and allow more background noise, alongside the work of Tan et al. that overcame the problem of protocols not being supported by finite-key proofs to allow experimental testing, as well as Liu et al.'s contributions to performing realistic testing to find key rates that support DIQKD over longer distances.

I. INTRODUCTION

Quantum Key Distribution (QKD) is a key distribution method that relies on properties of quantum mechanics to provide provably secure keys, rather than on computational complexity as in classical methods [1]. Hard problems have been the foundation of modern cryptographic protocols traditionally, but with the looming possibility of a quantum computer able to crack protocols like Diffie-Hellman, researchers have started looking for an alternative key-exchange protocol resistant to such attacks. QKD ensures security since if an attacker attempts to intercept quantum communication, their measurement will collapse the quantum state, introducing detectable anomalies in the transmission that would alert the parties to an intrusion in their communication.

However, standard QKD methods, such as the original BB84, are difficult to implement in practice and vulnerable to side-channel attacks among other attack vectors [2]. QKD also relies on the two quantum communication devices to behave exactly as defined by the protocols. In practice, verifying the characterization of each quantum state throughout the execution of the protocol has proven difficult to implement and is vulnerable to exploitation. An attack as simple as using a laser to shoot into the communication line can increase the

intensities of the prepared states, thereby compromising the information-theoretical security [1].

Device-Independent Quantum Key Distribution (DIQKD) alleviates these problems by relying solely on observed nonlocal correlations certified by Bell inequality violations, without requiring assumptions about the internal workings of the measurement devices [2]. Figuratively speaking, DIQKD assumes that each measurement device is a black box rather than working directly with the devices as BB84 does. With the assumptions of each measurement device removed, vulnerabilities arising from quantum state management from side-channel attacks by adversaries are mitigated as well as removing the problems arising from competing quantum communication devices from alternate vendors.

In this paper, we will review the initial landscape of DIQKD from 2021 onward, covering theoretical and experimental tests, with a particular focus on a high-level overview of its foundational concepts. Additionally, we will explore the evolution of several research methods from proofs of concept to more recent lab successes and examine challenges and future research directions.

II. BACKGROUND

A. Bell Inequalities and the CHSH Game

The Clauser-Horne-Shimony-Holt (CHSH) test is the foundation of why DIQKD works [3]. The test operates as follows: Suppose Alice and Bob are playing a game in which they are each given a random bit $X, Y \in \{0, 1\}$ by a referee independently. During the round, they must take a measurement and communicate their result $A, B \in \{0, 1\}$ back to the referee and try to submit an answer so that $A \oplus B = X \cdot Y$. However, a rule prohibits Alice and Bob from communicating once the round starts. If each possible outcome were tested, we would find that, in classical terms, the highest probability is 75%, or 3 out of 4 rounds in which they win. However, if Alice and Bob try to maximize their odds of winning, they can use the correlation between shared entangled quantum particles to coordinate their outputs. If Alice measures her entangled particle and gets 1, then she knows Bob will get 0, and vice versa. And since they will be able to determine each others output, they will always output $A, B \in \{0\}$ to the referee. Using this method, we instead find that the probability of winning increases to approximately 85.4% [3].

The fact that we can exceed the 75% win rate violates Bell's inequality, which means the result had to have come from an entangled state and can therefore be used as a secure state. Additionally, this introduces the possibility of a black-box source, as we do not need to make any assumptions about the devices as long as the test result is a violation. The closer to the threshold of 85.4%, the stronger the security of the communication is, as it further solidifies an entangled-state source. However, if the result is less than 75%, the source's legitimacy cannot be assumed, and the communication will be terminated.

B. DIQKD Security Model

The security of DIQKD relies entirely on continuous testing of Bell's inequality to assess transmission security. It makes no assumptions about the devices' internal workings. This allows the devices to be treated as a "black box," as there is no need to understand what they are doing to maintain the entangled state. However, DIQKD does require three assumptions about the communication itself to work:

- 1) Alice and Bob still require a classical channel to communicate their entangled quantum state.
- 2) A fully trusted random number generator.
- 3) Secrecy of Alice and Bob's location so that information about their inputs is hidden [3].

C. Determining Key Rate

Since no assumptions can be made about the quantum devices contained in the black box, determining the key rate (amount of usable raw data) relies on estimating the bounds of the observables after a certain number of rounds. These observables include S , the strength of the Bell violations, and the quantum bit error rate (QBER) of Alice and Bob's communication. In the presence of eavesdropper Eve, S is also a representation of how much information can be known by Eve. The larger the value of S is measured, the more uncertain Eve is. However, if S drops too low, Eve will gain an advantage in key derivation, and if QBER becomes too high, the error rate will make communication too inefficient to be effective.

D. Attack Vectors

The two main attack methods that Eve can use to eavesdrop on Alice and Bob's secret key are collective and coherent attacks. In a collective attack, eavesdropper Eve applies her quantum operation to each signal independently, then measures them jointly to obtain the key. In general, collective attacks are tested first, as they can provide a foundation for theoretical security before moving on to a more robust attack simulation. Coherent attacks differ as Eve is given the power to entangle her probe across all signals, which she can use to create a correlation between rounds, and it is much more difficult to defend against. Security of the protocol depends not only on defending against collective attacks, but also on defending against coherent attacks with positive key rates.

E. Entropy Accumulation

The Entropy Accumulation Theorem (EAT) provides a method to defend against coherent attacks at finite key sizes. Rather than measuring Eve's uncertainty across all signals, EAT allows for bounding of Eve's knowledge by computing how much of the secret key can be extracted after n rounds. As long as the extracted key remains within the bounds, the protocol's security is certified against coherent attacks.

III. IMPROVEMENTS IN DIQKD KEY EXTRACTION

While the initial proposals of DIQKD have been successful in lab environments, the practicality of their implementation remains a challenge. Outside of a controlled lab environment, noise levels are much higher, resulting in the key rate falling to unusable levels. To mitigate higher noise levels and achieve better key rates, researchers have been testing iterations of the original DIQKD protocol to find a method that can be implemented in real-world communication systems.

In 2021, Schwonnek et al. introduced an improvement of the original DIQKD protocol by using two randomly chosen key-generating bases instead of one. While the majority of the protocol functions the same as the original method, Bob is now given an additional measurement parameter, and the secret key is generated from both of Alice's measurements [4]. By randomly switching between the two bases, Eve's uncertainty increases as she cannot know in advance which basis is being used in the round. Using a depolarization model to determine the worst-case key rate, critical CHSH scores can decrease from 2.423 to 2.362 [4]. With the critical CHSH value lowered, the protocol can now tolerate higher levels of noise, increasing the QBER from 0.071 to 0.082 [4]. This improvement brings the practical implementation of DIQKD within the reach of additional experiments. Additionally, their protocol's tested security against coherent attacks indicates that approximately 10^8 – 10^{10} rounds are needed to achieve a positive key rate.

While Schwonnek et al. provided a theoretical improvement upon the original DIQKD method, it lacked a finite-key proof for improved protocol variants. All previous iterations of DIQKD relied on running an infinite number of rounds, an operation that does not translate into practical implementation. Tan et al. alleviated this problem by publishing a proof for DIQKD that leans on EAT, which states that Eve's uncertainty, also known as von Neumann entropy, can be bound through analyzing the accumulation of entropy every round [5]. By imposing a bound on S , the proof guarantees the security of a key rate against depolarizing noise up to 0.0933, a major increase from 0.082 in the random-basis implementation of previous iterations [5].

Building on these theoretical advances, Liu et al. in 2022 adopted an alternative approach to mitigating noise tolerance in DIQKD, combining several methods, including random post-selection, noisy preprocessing, and von Neumann entropy estimation through their own developed numerical methods [6]. The researchers then tested the detection efficiency of their measurements by running experiments under more realistic conditions. While their iteration of the DIQKD

protocol did not aim to determine the QBER threshold as in previous theoretical research, their experiments instead found a usable key-value detection efficiency of $\sim 87.5\%$ over a distance up to 220 m. Such a high success rate indicates that the protocol implementation yields a large proportion of post-selected events suitable for key generation, representing a major step towards realistic DIQKD experiments that would pave the way for future long-range tests.

IV. OPEN PROBLEMS

Despite the year-over-year improvements in DIQKD testing in the early 2020s, several challenges remain to this day that must be overcome for the protocols to be practical. DIQKD implementation in real-world settings with increased background noise raises the threshold for maintaining positive key rates above current experimental results. Usable key rates within these noisy environments require more rounds than be currently implemented with quantum hardware. Additionally, as the distance over the quantum channel increases, the key rate will drop below the threshold needed for secure communications. While Liu et al. did implement realistic tests with detection efficiencies of 87.5% up to 220 m, their tests were still conducted within a controlled lab environment. The distances they achieved show that DIQKD research is progressing, and the later research in 2026 with distances up to 100 km signify an even greater leap, DIQKD still has a long way to go before it can be useful. Also, these research methods support only communication between two endpoints using unique fiber lines and are not compatible with the current communication infrastructure, which relies on numerous interconnected devices to route traffic. Realistic implementations would require DIQKD to be compatible with modern fiber or to use its own quantum routing points to connect distant devices. Both of which remain major scientific and economic challenges to overcome.

V. CONCLUSION

Overall, research on mitigating the effects of background noise during DIQKD communication has made significant progress in a short time, iterating quickly on prior theoretical work and moving into real-world experimentation. While the random-key basis of Schwonnek et al. improved key rates over the original DIQKD protocol, it lacked a finite-key proof [4]. Tan et al. addressed this problem by using entropy accumulation to provide the missing proof that determines the bounds of Eve's knowledge, thereby guaranteeing security for the improved protocols in more realistic levels of noise [5]. At the same time, Liu et al. approached the problem of noise from a more realistic experimental perspective, finding that through random post-selection and noisy preprocessing DIQKD can achieve sufficient event detection to demonstrate positive key rates over a longer distance. Together, these works indicate that research on DIQKD advances on both the theoretical and experimental fronts, building on each other towards possible implementations for communications outside controlled environments. However, numerous challenges still

lay in the path of DIQKD that must be overcome for the protocol to become implemented in current infrastructure, both from a protocol design and hardware perspective. Despite these challenges, researches have more recently made significant jumps, such as in February 2026 a team was able to increase the range DIQKD up to ranges of 100 km, signifying an exponentiation growth in advancement. [7].

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